Multi-head attention:-

"The man saw the astronomer with a telescope".

Its ambigous sentence with 2 meanings

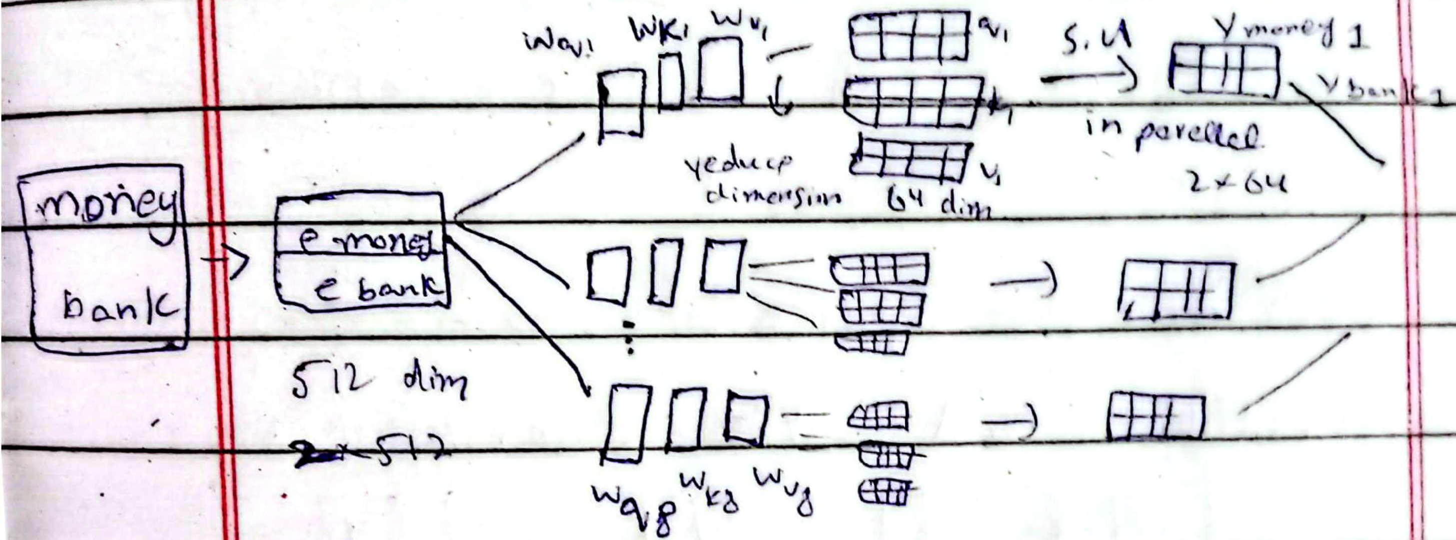
- Unfortunately self attention fails here as it figure out single perspective.

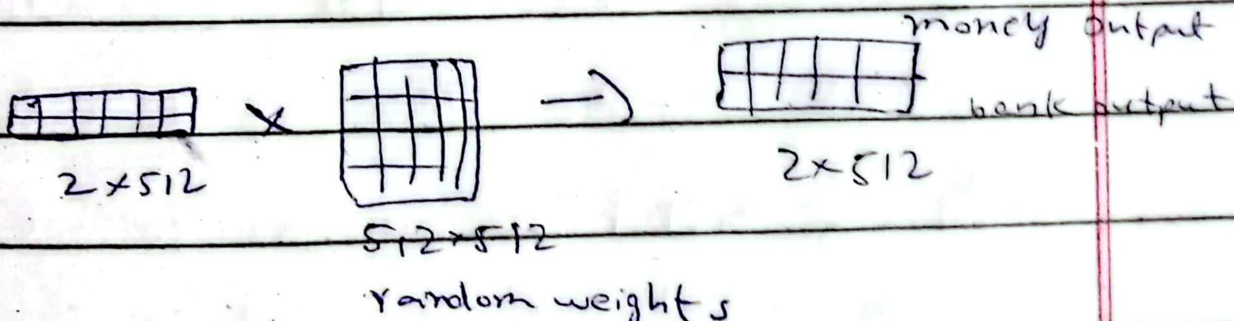
→ So the solution is multi head attention.

2 self attention → apply parallel.

Date: Day:

→ Use multiple self attention instead of 1 & for every self attention we call it a head, so that's why it is called multi-head attention.





So problem is solved in same computation of 1 self attention.

Date:________ Day:________

Positional encoding:-

Nitish killed lion
Lion killed Nitish. | goes in parallel to 1 another

→ If u pass sentences like this to self attention, they will say they r same, & that is a major drawback.

1st principle approach:-

Nitish killed the lion.

[| | | 1] [| | | 2] [| | | 3] [| | | 4]

- but think if we have pdf of 1 lac words, & N.N hate big nos so fails.

→ So we need a bounded, continuous & periodic function.

What if we use positional encoding → Sin func
but here problem is periodicity. (no not unique for all)
introduce Cos (Pos) & then gradually increase dimensions.

We don't concatenate this to embedding, we add these dimensions to each dimension of embedding

Date: ______ embedding vect + positional encoding vector. Day: ______

Note:- Position encoding will be a vector for every word.

→ Dimension will be exactly same like embedding vector.

→ We will add pos + emb $\xrightarrow[\text{resultant to}]{\text{send}}$ Self attention.

Layer normalization:-

Normalization in dl applied to input layer & hidden layer.

→ Process of transforming data or model outputs to have specific statistical properties, typically a mean of 0 & variance of 1.

→ Batch normalization cannot be used in transforme due to more zero's of padding such as 1st sentence → 100 words
other " → 30 words

batch — features.

Date: Day:

We find out layer normalization instead of batch.

→ In layer normalization we apply across features.

hence σ's impact ↓.